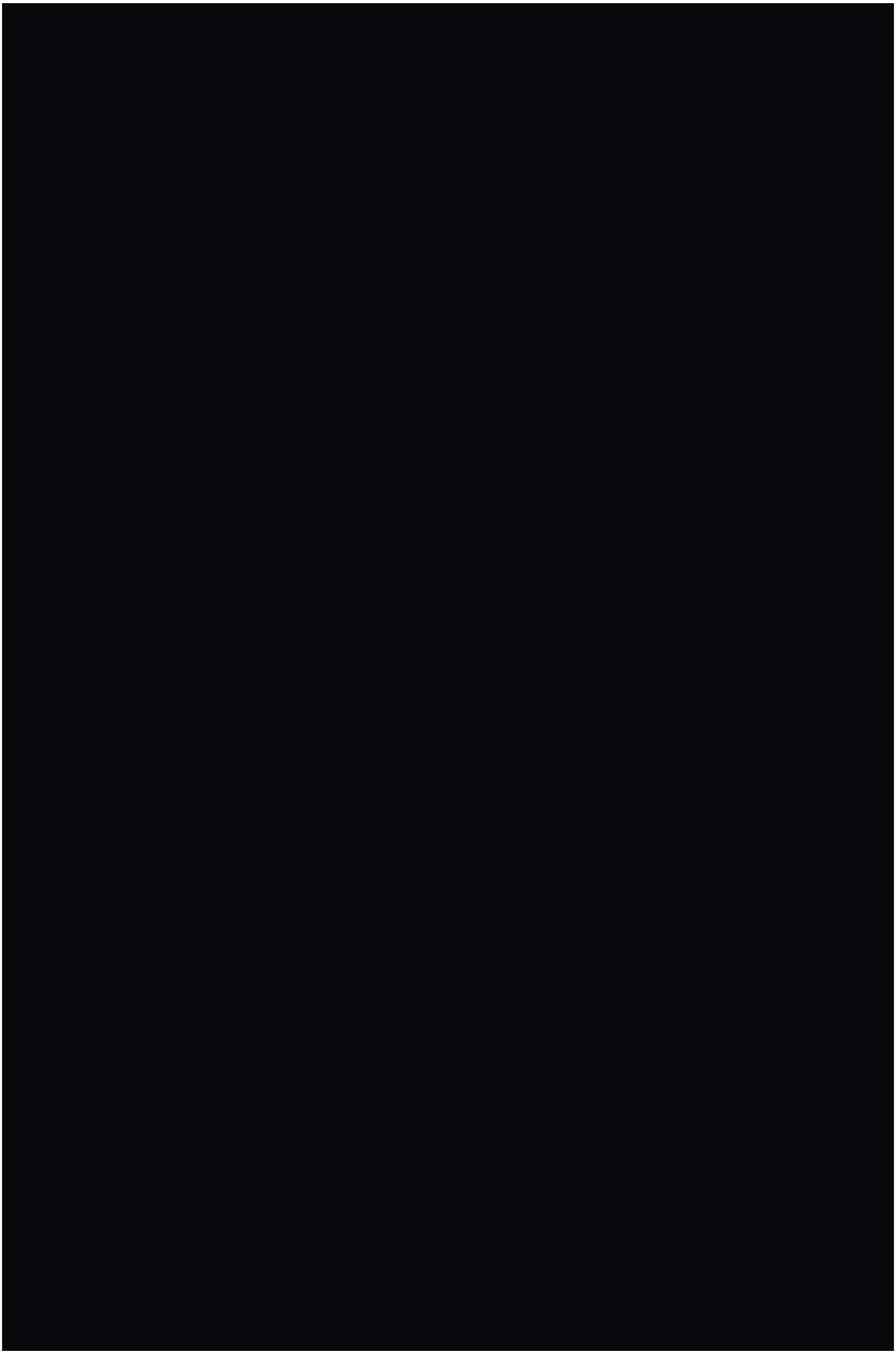
REPAIRO // SYSTEM

The Self-Healing Code Engine for API Dependencies

Sequoia & McKinsey-Style Strategic Investment Memorandum

PREPARED FOR: SANJAY NANDA N J & LEAD INVESTORS
AUTHOR: ADITYA (FOUNDER & CEO)
DATE: AUGUST 2026

ENTITY: DELAWARE C-CORP (TOPCO) / INDIAN PVT LTD (OPCO)
STATUS: PRE-SEED / FUNCTIONAL CORE BETA MVP
BACKING: SARVAM STARTUP PROGRAM



1. EXECUTIVE SUMMARY

In modern cloud software, third-party APIs represent the primary attack surface for integration failure. Monoliths have fragmented into microservices, and AI integrations (LLMs, vector stores, agentic routing) are evolving at a chaotic pace. However, the developer tooling surrounding API dependencies remains entirely diagnostic—alerting engineers of failure without offering remediation.

Repairo is the world's first autonomous self-healing code engine for API dependencies. When a vendor (such as Stripe or OpenAI) deprecates parameters, renames fields, or updates endpoints, Repairo immediately parses the downstream client's repositories into an Abstract Syntax Tree (AST), maps call sites, and outputs deterministic, review-ready Pull Requests to fix the code.

INVESTMENT THESIS WEDGE:

By transitioning the developer workflow from *passive observability (alerts)* to *active remediation (PR fixes)*, Repairo eliminates 40% of microservice integration failures and saves engineering teams hundreds of manual maintenance hours annually.

2. MARKET OPPORTUNITY (TAM/SAM/SOM)

The global API economy has expanded exponentially, with the average enterprise maintaining over 15,000 active API connections. This has created a massive, unmanaged code maintenance overhead.

Market Size Projections (2026 - 2034)

- **Total Addressable Market (TAM): $22.4 Billion**
  The total global API Development and Management Tools market size projected by 2034, driven by the expansion of SaaS endpoints and microservice architectures.
- **Serviceable Addressable Market (SAM): $4.5 Billion**
  The developer operations, security compliance, and code integration segment targeting mid-market and enterprise B2B software architectures.
- **Serviceable Obtainable Market (SOM): $350 Million**
  The initial beachhead segment targeting high-velocity, AI-native startups and devtool companies deploying to next-gen edge architectures (Vercel, AWS Lambda, Cloudflare Workers).

>15,000

ACTIVE API ENDPOINTS PER ENTERPRISE

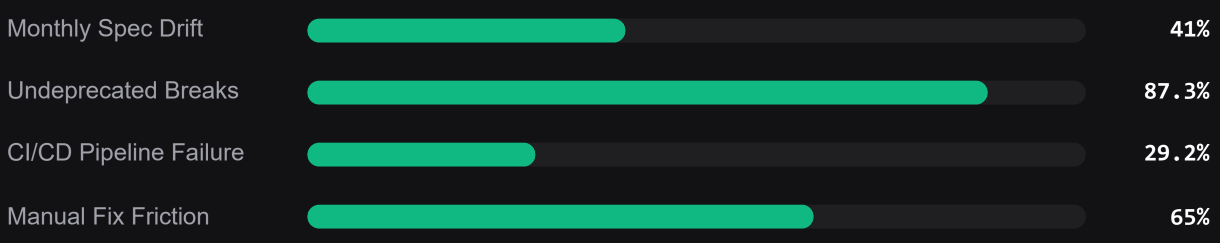
$200,000

AVG. COST PER HOUR OF API DOWNTIME

3. PROBLEM VALIDATION & RESEARCH METRICS

The traditional approach of manually auditing changelogs and codebases fails at scale. Quantitative research across millions of lines of production code exposes systemic contract fragility:

Systemic API Fragility Indicators



- **KushoAI 2026 Spec Drift:** 41% of third-party APIs experience schema drift within 30 days, rising to 63% within 90 days.
- **The RADA Spec Study:** Out of 2,224 OpenAPI specs evaluated, 87.3% of breaking versions introduced changes silently without prior deprecation warnings.
- **Developer Productivity Drain:** Engineers waste an average of 3 hours per week (20 full workdays per year) diagnosing and refactoring third-party API breakages.

4. TECHNICAL ARCHITECTURE

Repairo utilizes a secure, low-latency, deterministic transformation pipeline that avoids standard LLM hallucinations and data exfiltration concerns.

```
[OpenAPI spec change feed] ──► [Spec Differ (Detects change)]
 |
 ▼
[Customer Codebase] ──► [Volatile RAM Vault (~24ms)]
 |
 ▼
[GitHub App PR] ◄── [Deterministic AST Codemod (Tree-sitter)]
```

Core Security Moat: 24ms Volatile RAM Vault

To pass Infosec reviews for enterprise clients (such as financial institutions or healthcare platforms):

- Client code is streamed directly into an ephemeral, isolated memory block (RAM).
- AST parsing and refactoring occur in-memory, the patch is committed, and the RAM state is completely zeroed out in **~24ms**.
- **Zero Disk Persistence:** No customer source code is ever written to disk, cache, database, or vector stores.

5. COMPETITIVE LANDSCAPE

Existing developer tools focus entirely on alerting, testing, or API documentation. Repairo is the only player offering automated code remediation.

Company	Primary Focus	Remediation Capability	Limitation
Postman / Akita	API monitoring & discovery	None (Alerting only)	Downstream team must fix code manually.
Speakeasy	SDK generation & spec linting	SDK re-generation only	Does not patch consumer client logic.
Pact / PactFlow	Contract testing	None (Build block)	Useless for third-party endpoints.
Dependabot	Package dependency updates	Version bumps	Ignorant of code logic; causes compile breaks.
Repairo	**Drift Detection + AST PRs**	**Full AST PRs**	**Self-healing remediation in 24ms.**

6. TWO-YEAR EXECUTION ROADMAP

Q1 (Months 1-3): The Foundation & Core Engine

Complete core AST parsing for TypeScript/Node.js. Incorporate Delaware TopCo + Indian OpCo. Close $100k SAFE note.

Q2 (Months 4-6): Integration & Closed Alpha

Build spec poller. Onboard 3-5 unpaid design partners. Hire 1x Senior Backend Engineer.

Q3 (Months 7-9): Public Beta & SOC2 Audit

Launch self-serve GitHub App. Initiate SOC2 Type I compliance audit. Reach 50 active installed repos.

Q4 (Months 10-12): Monetization & YC Entry

Expand support to Python. Implement Stripe billing ($99/mo). Target Y-Combinator accelerator entry.

Q5 - Q8: Enterprise scale & Seed

Build GitLab/Bitbucket integrations. Scale to $30k+ MRR. Raise $1.5M Venture Seed Round. Transition team to full salaries.

7. GO-TO-MARKET & FINANCIAL PROJECTIONS

Our GTM is purely developer-focused and leverages product-led growth (PLG) alongside targeted content marketing.

Ideal Customer Profile (ICP)

- **Wedge:** Series A/B startups heavily consuming volatile Generative AI APIs in production TypeScript and Python environments.
- **Expansion:** Mid-market enterprise engineering teams (200+ developers) with dense microservice architectures.

Financial Projections (24-Month Target)

Metric	Q1	Q4	Q6	Q8 (Seed Close)
Monthly Recurring Revenue	$0	$3,000	$15,000	$30,000+
Active Connected Repos	5	50	150	250+
Full-Time Headcount	2	4	6	9

8. FUNDRAISING & INVESTMENT THESIS

The Funding Ask: $250k Pre-Seed SAFE

We are opening a **$250,000 Pre-Seed SAFE note** at a **$3.5M post-money valuation cap** to fund the next 18 months of execution. Capital allocation will be directed toward Delaware legal compliance, cloud infrastructure (Railway/AWS), and founder/engineering stipends.

Why Repairo Wins (Investment Thesis):

1. **Defensible Moat:** Parsing and refactoring code via AST codemods provides a highly precise, hallucination-free solution that standard LLM-completion wrappers cannot match.
2. **Enterprise Readiness:** The 24ms Volatile RAM Vault bypasses standard data-retention hurdles, allowing us to pass SOC2 and secure enterprise pilots from Day 1.
3. **Stellar Early Team:** A technical founder paired with an ex-BlackRock Systems Lead and specialized security/compiler leads guarantees elite, rapid execution.